

NK2R signaling governs intestinal lipid mobilization and mucosal inflammation

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eLife Assessment

This is an **important** study on the role of the neurokinin-2 receptor (NK2R) as a regulatory node connecting intestinal lipid metabolism, mucosal immunity, and the gut microbiome, bidirectionally regulating enterocyte lipid uptake, lipid droplet storage, chylomicron output, and systemic metabolic parameters in DIO mice. The authors present **solid** evidence linking *Tacr2* deletion to reprogrammed epithelial lineage allocation, dampened immune gene expression, and male-biased protection from DSS colitis, despite dysbiotic microbiota. However, the causal evidence for some mechanistic and pro-inflammatory NK2R claims remains **incomplete** and potentially confounding, requiring additional cell-type-specific and functional experiments.

Abstract Neuropeptidergic control of lipid metabolism is conserved and increasingly implicated in metabolic diseases, but receptor-level mechanisms remain unclear. Here, we identify the neurokinin-2 receptor (NK2R) as a central node linking tachykinin signals to intestinal lipid mobilization, epithelial composition, and mucosal inflammation. Across complementary genetic and pharmacological perturbations, modulation of NK2R drives bidirectional effects. Loss or blockade of NK2R increases postprandial triglyceridemia and expands intestinal lipid stores, whereas agonism suppresses chylomicron output, reduces adiposity, and improves glycemia in diet-induced obesity. Transcriptomic and cellular analyses indicate coordinated upregulation of lipid-metabolic programs with a concomitant dampening of immune pathways in the absence of NK2R, accompanied by sex-specific remodeling of secretory lineages and male-biased protection from colitis. NK2R signaling also shaped the fecal microbiota in a genotype- and diet-dependent manner, highlighting crosstalk among neuropeptide signaling, epithelial physiology, and host-microbe interactions. These findings position NK2R as a molecular switch for intestinal lipid handling and mucosal inflammation and suggest that NK2R-targeted agonists or antagonists could be deployed as context- and sex-dependent therapeutic strategies for metabolic disease and inflammatory bowel disease.

Introduction

Neuropeptidergic control of lipid metabolism is increasingly recognized as a conserved feature of energy homeostasis (Yoshinari et al., 2021; Palamiuc et al., 2017; Song et al., 2014; Kuo et al., 2007; Park et al., 2017; Yin et al., 2015; Hou et al., 2021). Using the *Caenorhabditis elegans*

model system, we had previously delineated a brain-to-gut neuroendocrine axis in which the sensory neuron-derived neuropeptide FLP-7 activates the intestinal G protein-coupled receptor NPR-22 to stimulate fat mobilization via induction of the adipocyte triglyceride lipase ATGL-1 and trigger fat loss via mitochondrial β -oxidation and increased energy expenditure (Palamiuc et al., 2017; Littlejohn et al., 2020). Phylogenetic analyses show that FLP-7 sequence resembles the mammalian tachykinin peptide family, and NPR-22 is the *C. elegans* ortholog of the mammalian neurokinin-2 receptor (NK2R) (Palamiuc et al., 2017). We have established NPR-22 as both necessary and sufficient for driving fat loss in the worm intestine (Palamiuc et al., 2017). Taken together, these findings are consistent with a conserved mammalian tachykinin-NK2R axis controlling lipid mobilization, prompting our examination of mammalian tachykinins and their receptors.

The mammalian tachykinins, including substance P, neurokinin A, and neurokinin B, constitute a conserved neuropeptide family that has historically been known to mediate nociception, inflammation, cancer progression, and gastrointestinal functions (Nässel et al., 2019; Suvas, 2017; Steinhoff et al., 2014; Ziegglänsberger, 2019; Onaga, 2014). These ligands signal through three G protein-coupled receptors (GPCRs): neurokinin-1 receptor (NK1R), neurokinin-2 receptor (NK2R), and neurokinin-3 receptor (NK3R), which are preferentially activated by substance P, neurokinin A, and neurokinin B, respectively. Prior studies have established NK1R as a key regulator of pain, inflammation, and immune responses (Hegron et al., 2023; Mantyh, 1991; Mishra and Lal, 2021; Martinez and Philipp, 2016). NK1R antagonists have demonstrated therapeutic potential by attenuating blood-brain barrier dysfunction, cerebral edema, and pro-inflammatory cytokine levels in traumatic brain injury models (Li et al., 2019; Vink and van den Heuvel, 2010; Donkin et al., 2009; Gabrielian et al., 2013). In contrast, NK3R has been extensively studied in reproductive biology and neurological disorders (Steinhoff et al., 2014; Zhang et al., 2020; Navarro et al., 2015). The neurokinin B-NK3R signaling is critical for hypothalamic-pituitary-gonadal axis function, with an NK3R antagonist now clinically deployed to relieve menopausal vasomotor symptoms (Lederman et al., 2023; Depypere et al., 2021). Beyond reproduction, NK3R has also been implicated in addiction, pain, and psychiatric disorders (Zhang et al., 2020; Schank, 2020; Zhang et al., 2023; Hether et al., 2013).

By comparison, NK2R regulates diverse physiological processes, including smooth muscle contraction, nociception, intestinal fluid secretion, and reproductive functions (Nakamura et al., 2011; Meuchel et al., 2011; Lecci et al., 2004; Delvalle et al., 2018; Torres et al., 2021; León et al., 2019). Although NK2R is expressed in multiple tissues, including the gastrointestinal tract, lung, and immune system (Mapp et al., 2000; Kitamura et al., 2012; Ohtake et al., 2015; Vannucchi and Faussone-Pellegrini, 2000; Holzer and Holzer-Petsche, 2001; Lecci et al., 2006; Renzi et al., 2000), its specific roles in metabolism and immunity remain poorly defined. A recent study (Sass et al., 2024) reported that the administration of a selective NK2R peptide agonist increases energy expenditure, reduces fat mass, and suppresses appetite, highlighting its therapeutic potential for cardiometabolic diseases, including obesity and diabetes. However, the molecular mechanisms engaged by NK2R agonism are largely unknown.

Results

Expression pattern of Tachykinin Receptor 2 (*Tacr2*)

Our prior work in *C. elegans* showed that the CeNK2R gene *npr-22* is expressed in a few pairs of head neurons and in the intestine, but not in the muscle or hypodermal tissues (Palamiuc et al., 2017; Figure 1A). We examined *Tacr2* expression in mammalian species: the Human Protein Atlas (Karlsson et al., 2021; Uhlén et al., 2015) RNA-seq datasets revealed that human *TACR2* is preferentially co-expressed with genes enriched in the intestinal 'digestion' cluster (Figure 1B). In contrast, human *TACR1* and *TACR3* clustered with transcripts characteristic of connective tissue and retina, respectively (Figure 1—figure supplement 1A). Consistent with these patterns, quantitative PCR across mouse small-intestinal segments showed that *Tacr2* mRNA levels exceeded those of *Tacr1* and *Tacr3* along the entire proximal-distal axis (Figure 1C). Immunofluorescence localized the NK2R protein to the basolateral membrane of mouse intestinal epithelial cells (Figure 1D), indicating a potential role in responsiveness to internal physiological signals rather than luminal nutrients. These results are consistent with our previous findings in *C. elegans* (Palamiuc et al., 2017), in which NPR-22, the

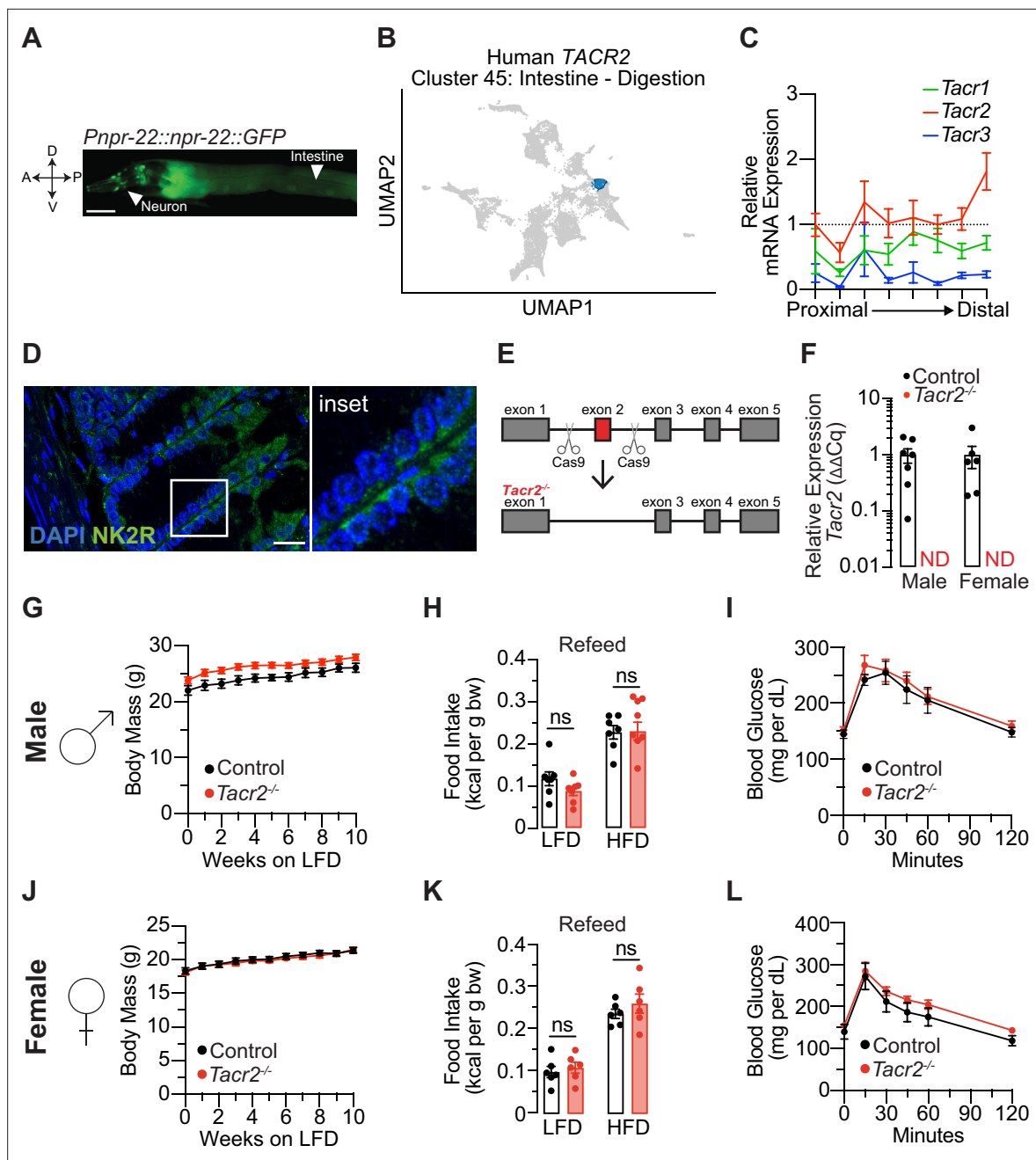


Figure 1. Tachykinin receptor 2 tissue expression. **(A)** Fluorescent image of a transgenic *C. elegans* expressing *npr-22::GFP* under the control of the endogenous *npr-22* promoter. GFP expression was detected in the intestinal cells and in several pairs of neurons in the head. A, anterior; P, posterior; V, ventral; D, dorsal. Scale bar, 50 μ m. This panel was reproduced from Figure 7B of *Palamiuc et al., 2017*. **(B)** UMAP visualization depicting gene clusters based on mRNA expression profiles from The Human Protein Atlas (*Uhlén et al., 2015*) (HPA). The cluster containing human *Tacr2* (cluster 45, comprising 300 genes) is highlighted in blue. Functional annotation provided by HPA characterizes the shared specificity (Intestine) and biological function (Digestion) of genes within this cluster. **(C)** The gene expression levels of *Tacr1*, *Tacr2*, and *Tacr3* were assessed throughout the entire mouse small intestine using qPCR. The intestine was divided into eight equal segments, each measuring approximately 4–5 cm. *Actb* was used as the housekeeping gene. Data are presented as the fold change relative to the expression level of *Tacr2* in the first segment $\pm$ SEM. $n=6$ biological replicates. **(D)** Fluorescent image of mouse jejunum stained with neurokinin-2 receptor (NK2R) antibody (green) and DAPI (blue). Scale bar: 20 μ m. Inset: magnified view of the region enclosed by the white line in the fluorescent image. The jejunum was harvested from male C57BL6/J mice (8 weeks old, low-fat diet). **(E)** The strategy of CRISPR-Cas9 mediated genome editing, depicting the genomic region of *Tacr2* and the locations of the Cas9 cutting sites for *Tacr2*^{-/-}. The deleted exon 2 is marked in red. **(F)** qPCR analysis of *Tacr2* mRNA expression in intestinal epithelial RNA isolated from male and female control and *Tacr2*^{-/-} mice. $n=7$ male control, $n=6$ female control, $n=6$ male *Tacr2*^{-/-}, $n=6$ female *Tacr2*^{-/-}. **(G, J)** Body weights of male (**G**) and female (**J**) *Tacr2*^{-/-} and control mice (8–12 weeks old) were monitored weekly over a 10 week period while maintained on a low-fat diet (LFD). $n=8$ control and

Figure 1 continued on next page

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n=12 *Tacr2*^{-/-} mice (G); n=11 control and n=13 *Tacr2*^{-/-} mice (J). (H, K) Following an overnight fast, male (H) and female (K) mice were re-fed with either LFD or high-fat diet (HFD), and cumulative food intake was measured over a 4 hr refeeding period. n=7 control (LFD), n=7 *Tacr2*^{-/-} (LFD), n=7 control (HFD), and n=8 *Tacr2*^{-/-} (HFD) mice (H); n=6 per group (K). (I, L), Glucose tolerance tests were performed in week 10 in male (I) and female (L) mice. n=5 control and n=6 *Tacr2*^{-/-} mice per group. Data was analyzed using an unpaired Student's t-test (two-tailed) (F, H, K) or two-way ANOVA with repeated measures (G, I, J, L). Data are shown as mean ± SEM. *p<0.05. Black bars/lines represent control mice; red bars/lines represent *Tacr2*^{-/-} mice.

The online version of this article includes the following figure supplement(s) for figure 1:

Figure supplement 1. *Tacr1* and *Tacr3* expression in *Tacr2*^{-/-} male mice.

invertebrate ortholog of NK2R, is similarly expressed in intestinal cells and responds to internal signals from sensory neurons.

Generation of *Tacr2*-null mice and analysis of gene expression in response to dietary challenges

To investigate NK2R function in vivo, we generated global *Tacr2*-null (*Tacr2*^{-/-}) mice by CRISPR-Cas9 gene editing (Figure 1E). qPCR of jejunal epithelium confirmed loss of *Tacr2* transcripts (Figure 1F) without compensatory up-regulation of *Tacr1* or *Tacr3* (Figure 1—figure supplement 1B). *Tacr2*^{-/-} mice of both sexes were grossly normal, displaying wild-type body mass, re-feeding behavior, and glucose tolerance (Figure 1G–L).

Given the role of the *C. elegans* ortholog NPR-22 in intestinal lipid metabolism (Palamiuc et al., 2017), we wished to determine whether NK2R modulates mammalian intestinal lipid handling. *Tacr2*^{-/-} and wild-type littermates were maintained for one week on either a low-fat diet (LFD) or a Western-style high-fat diet (HFD) before jejunal epithelial RNA was profiled by RNA-seq (Figure 2A). Body mass remained comparable between *Tacr2*^{-/-} and wild-type littermates of both sexes on either LFD or HFD (Figure 2—figure supplement 1A, B).

In males, principal component analysis (PCA) revealed distinct clustering patterns: the wild-type mice clustered closely irrespective of diet; *Tacr2*^{-/-} mice fed a LFD formed a separate cluster from wild-types; *Tacr2*^{-/-} mice fed a HFD exhibited markedly different clustering compared to all other groups (Figure 2B). Only 221 genes were differentially expressed (DE) between genotypes on LFD, whereas 3030 genes were DE on HFD (Figure 2C and D), suggesting that loss of *Tacr2* selectively sensitizes the male intestinal epithelium to dietary fat. Interestingly, Gene Ontology (GO) enrichment linked up-regulated transcripts to lipid-metabolic pathways and down-regulated transcripts to immune and inflammatory processes (Figure 2E–H), concordant with prior *C. elegans* work that demonstrates a role for NK2R in intestinal lipid metabolism (Palamiuc et al., 2017; Littlejohn et al., 2020).

In females, the PCA was driven principally by genotype rather than diet (Figure 2I). The number of DE genes remained similar under LFD and HFD (Figure 2J and K), yet functional annotations mirrored those in males: lipid metabolism among up-regulated genes and immune response among down-regulated genes (Figure 2L–O). Overlap between diet-specific DE gene sets was modest in males but substantial in females (Figure 2P and Q), indicating a differential response in males to dietary fat in the absence of NK2R.

NK2R governs epithelial lineage allocation and intestinal inflammation

Interestingly, genes downregulated in *Tacr2*^{-/-} jejunum of both sexes were enriched for GO terms linked to immune response, immune regulation and leukocyte activation (Figure 2F, H, M and O), indicating an overall dampening of mucosal immunity. Bisque deconvolution (Jew et al., 1971) of bulk RNA-seq revealed a basal shift in epithelial lineages in male *Tacr2*^{-/-} mice (Figure 3A). Tuft and transit-amplifying (TA) cell signatures were elevated, whereas stem-cell and goblet-cell signatures were reduced; HFD further exacerbated these baseline genotype differences, notably by reducing the relative proportion of enterocytes. Paneth cell signatures declined after HFD in both genotypes, consistent with previous studies (Lee et al., 2017; Liu et al., 2021), while endocrine cell abundance was unchanged. In contrast, female mice showed only minor alterations (Figure 3—figure supplement 1A); in *Tacr2*^{-/-} females, HFD lowered stem-cell and enterocyte signatures and increased TA cells.

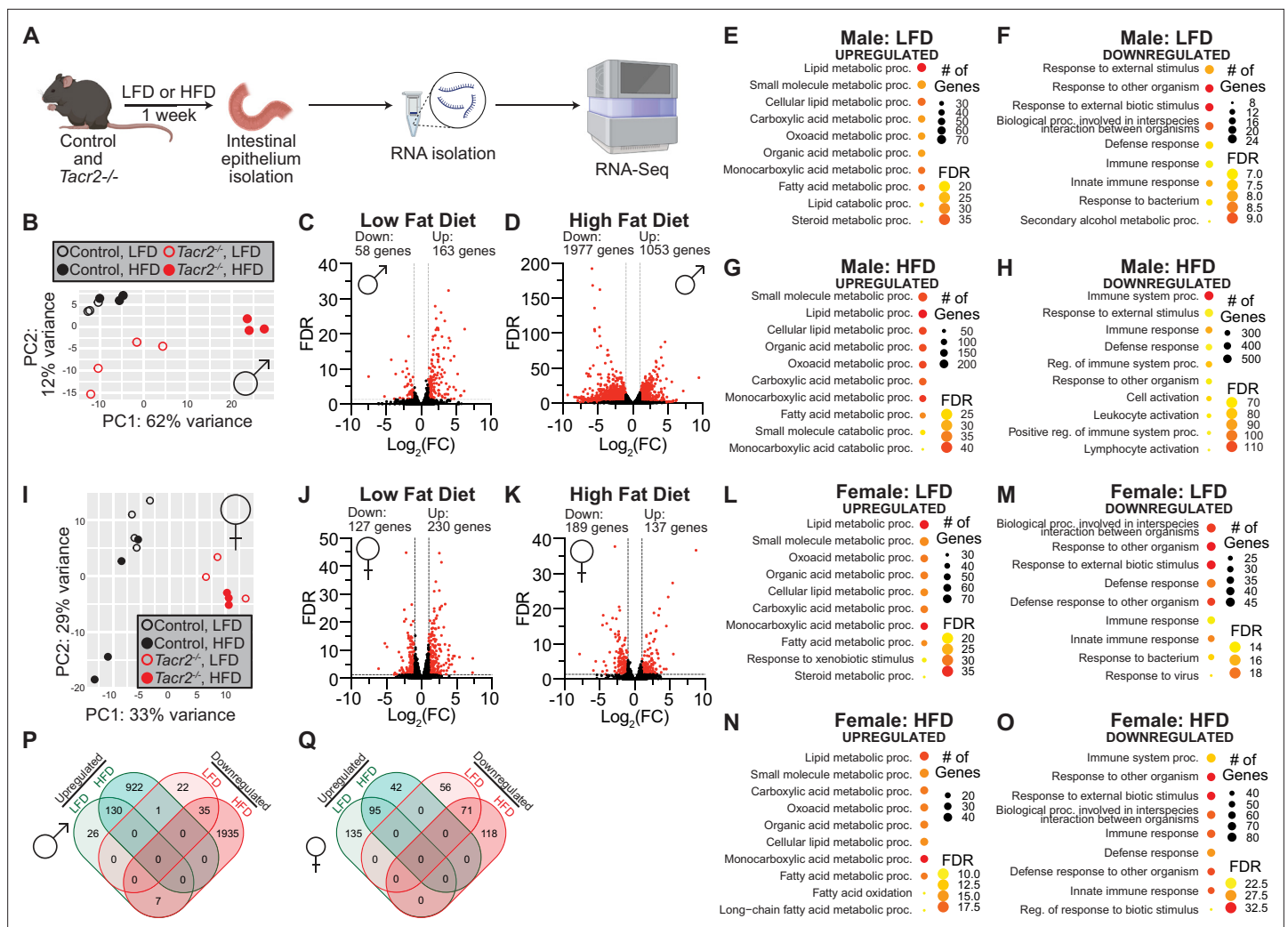


Figure 2. Sex-specific transcriptional responses to high-fat diet in *Tacr2*^{-/-} mice. **(A)** Schematic of the experimental workflow (created with BioRender.com). Control and *Tacr2*^{-/-} mice were fed a low-fat diet (LFD) or high-fat diet (HFD) for 1 week before sacrifice. Following tissue collection, the upper intestinal epithelial layer was isolated, and total RNA was extracted for subsequent bulk RNA sequencing. **(B)** Principal component analysis (PCA) of the RNA-Seq data from male control and *Tacr2*^{-/-} mice on LFD or HFD. *n*=3 control (LFD), *n*=3 control (HFD), *n*=4 *Tacr2*^{-/-} (LFD), *n*=3 *Tacr2*^{-/-} (HFD). **(C, D)** Volcano plots of differentially expressed genes in male *Tacr2*^{-/-} versus control mice under LFD **(C)** or HFD **(D)** conditions. **(E–H)** Gene ontology (GO) enrichment analyses of genes differentially expressed in male *Tacr2*^{-/-} mice versus controls on LFD **(E)**, upregulated; **(F)**, downregulated or HFD **(G)**, upregulated; **(H)**, downregulated. Bubble size indicates the number of genes associated with each GO term, and color represents the false discovery rate (FDR). **(I)**, Principal component analysis of RNA-Seq data from female control and *Tacr2*^{-/-} mice on LFD or HFD. *n*=4 control (LFD), *n*=4 control (HFD), *n*=3 *Tacr2*^{-/-} (LFD), *n*=3 *Tacr2*^{-/-} (HFD). **(J, K)** Volcano plots of differentially expressed genes in female *Tacr2*^{-/-} versus control mice under LFD **(J)** or HFD **(K)** conditions. **(L–O)** Gene ontology enrichment analyses of genes differentially expressed in female *Tacr2*^{-/-} mice versus controls on LFD **(L)**, upregulated; **(M)**, downregulated or HFD **(N)**, upregulated; **(O)**, downregulated. Bubble size and color reflect gene count and FDR, respectively. **(P, Q)** Venn diagrams showing overlap of differentially expressed genes (adjusted *p*-value < 0.05 and |log₂ fold change| > 1.0) between LFD and HFD conditions in male **(P)** and female **(Q)** datasets. Numbers indicate the number of shared or unique genes between conditions.

The online version of this article includes the following figure supplement(s) for figure 2:

Figure supplement 1. *Tacr2*^{-/-} mice show no significant changes in body weight compared to wild-type controls under acute high-fat diet (HFD) challenge.

Building on the transcriptional changes in immune-related genes observed in *Tacr2*^{-/-} mice, we next assessed functional consequences of *Tacr2* loss in vivo by inducing colitis in mice using the well-established dextran sodium sulfate (DSS) model (Chassaing et al., 2014; Figure 3B). Male *Tacr2*^{-/-} mice were protected from the deleterious effects of induced colitis, exhibiting attenuated weight loss, reduced plasma and fecal lipocalin-2 (LCN2) concentration (a marker of intestinal inflammation Chassaing et al., 2012), ameliorated histopathology with preserved architecture and less immune-cell

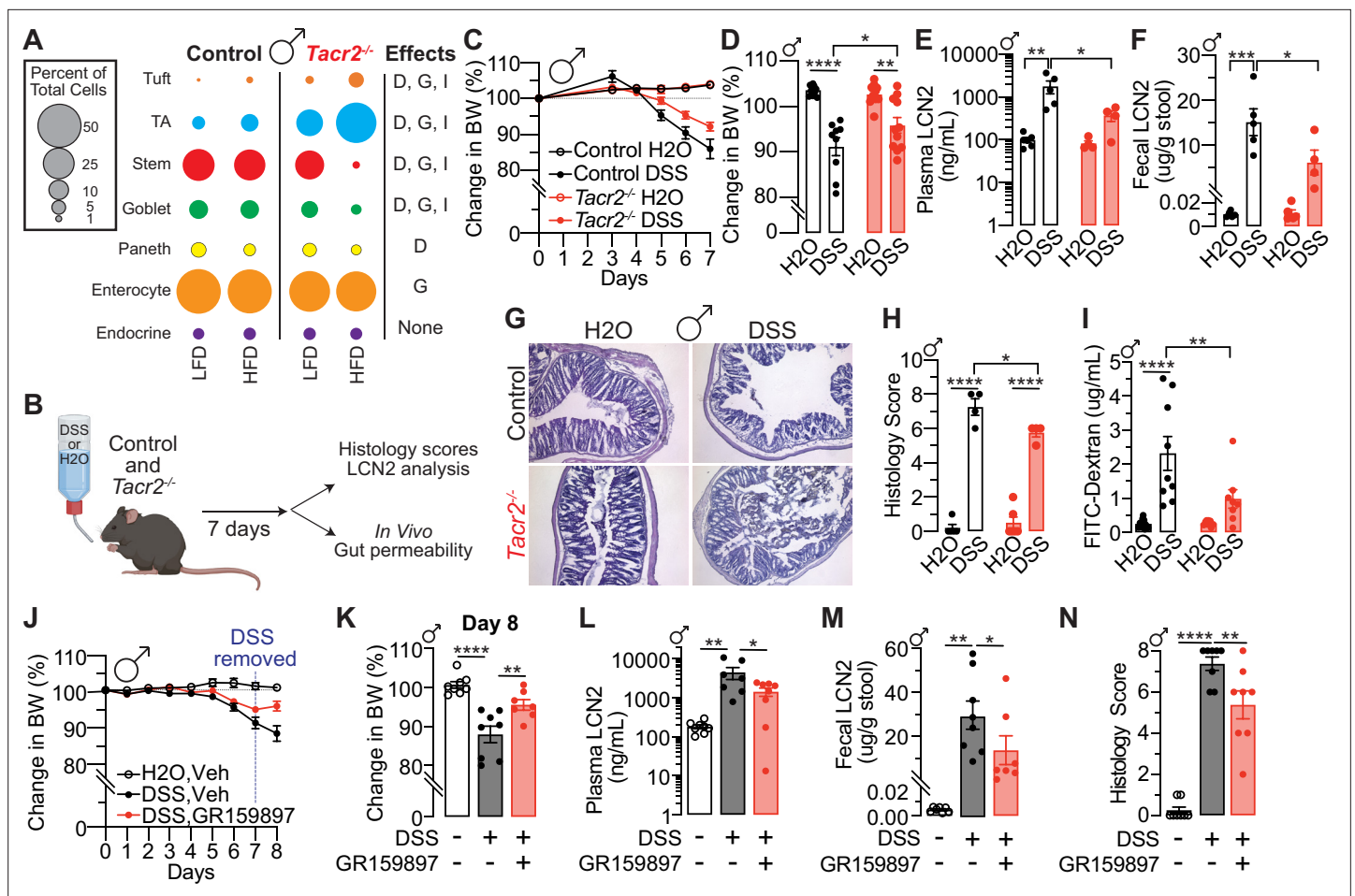


Figure 3. Neurokinin-2 receptor (NK2R) inhibition attenuates dextran sodium sulfate (DSS)-induced weight loss and intestinal inflammation in male mice. **(A)** Bulk RNA-Seq data were integrated with publicly available single-cell RNA-Seq datasets to estimate cell-type composition in the small intestine of male mice. Differences in estimated cell-type proportions were analyzed based on diet (D), genotype (G), or their interaction (I). **(B)** Schematic representation of the experimental workflow for the DSS-induced colitis model (created with BioRender.com). Male and female mice received DSS solution (3% for males, 5% for females) or vehicle (drinking water) ad libitum for seven consecutive days. Following treatment, tissues were collected for downstream analyses or subjected to an in vivo gut permeability assay. **(C)** Daily monitoring of body weights in male control and *Tacr2*^{-/-} mice treated with DSS or vehicle over a 7 day period. **(D)** Comparison of body weight changes from baseline (day 0) to day 7, derived from the data shown in panels (C). **(E, F)** Lipocalin-2 (LCN2) concentrations in blood (E) and feces (F) from male control and *Tacr2*^{-/-} mice after 7 days of DSS or vehicle treatment, as determined by ELISA. **(G)** Colon tissues from male control and *Tacr2*^{-/-} mice treated with DSS or vehicle for 7 days were collected, sectioned, and stained with hematoxylin and eosin (H&E). Representative images of stained colon sections are shown. **(H)** Representative histological images (G) were evaluated and scored according to the degree of tissue damage. **(I)** Gut permeability in male control and *Tacr2*^{-/-} mice following 7 days of DSS or vehicle treatment was assessed by oral gavage of 4 kDa FITC-dextran. Blood was collected 4 hr post-gavage, and FITC fluorescence was measured in plasma to quantify gut barrier integrity. **(J)** Daily monitoring of body weights in male wild-type mice treated with DSS or vehicle for 7 days, in combination with daily injections of the NK2R antagonist GR159897 (2.5 mg/kg) or vehicle (PBS). DSS administration was discontinued on day 7. **(K)** Comparison of body weight changes from baseline (day 0) to day 8, derived from the data shown in panel (J). **(L, M)** Lipocalin-2 (LCN2) concentrations in blood (L) and feces (M) from male wild-type mice with indicated treatment on day 8, as determined by ELISA. **(N)** Colon tissues from male wild-type mice with indicated treatment on day 8 were collected, sectioned, and stained with hematoxylin and eosin (H&E). Representative histological images were evaluated and scored according to the degree of tissue damage. Statistical analysis was performed using one-way or two-way ANOVA followed by Holm-Sidak post hoc tests. Data are shown as mean ± SEM, *****p*<0.0001, ****p*<0.001, ***p*<0.01, **p*<0.05. Black bars/lines represent control mice; red bars/lines represent *Tacr2*^{-/-} or GR159897-treated mice.

The online version of this article includes the following figure supplement(s) for figure 3:

Figure supplement 1. Neurokinin-2 receptor (NK2R) inhibition does not suppress dextran sodium sulfate (DSS)-induced weight loss and intestinal inflammation in female mice.

infiltration. In a second experiment performed under identical conditions, we observed improved epithelial barrier function indicated by reduced FITC-dextran permeability via the intestinal lumen (**Figure 3C–I**). In contrast, female *Tacr2*^{-/-} mice showed no protection in weight loss or plasma LCN2, increased fecal LCN2, and histological injury comparable to controls, although barrier integrity was partially maintained (**Figure 3—figure supplement 1B–H**). Of note, male *Tacr2*^{-/-} mice exhibited increased numbers of differentially expressed genes following HFD, an inflammatory insult (**Ma et al., 2008; van der Logt et al., 2013; Cheng et al., 2016; Teixeira et al., 2011**), and demonstrated colitis protection, whereas female *Tacr2*^{-/-} mice lacked this expanded gene expression response and did not exhibit protection against DSS-induced colitis. Together, these data indicate a sex-dependent transcriptional and functional responses to inflammatory challenge in the absence of NK2R.

Prompted by this result, we tested whether acute receptor antagonism via a selective NK2R antagonist would phenocopy the genetic absence, and help ascertain translational tractability. Single-dose pharmacological inhibition corroborated the genetic findings: daily administration of the selective NK2R antagonist GR159897 during DSS exposure mitigated weight loss, suppressed plasma and fecal LCN2, and improved histological scores in males (**Figure 3J–N**). Together, these data identify NK2R as a sex-dependent regulator of epithelial lineage allocation and susceptibility to mucosal injury.

Role of NK2R in Lipid Metabolism

Guided by our jejunal epithelial RNA-seq results, we next wished to determine the role of NK2R in lipid metabolism. Genes up-regulated in the *Tacr2*^{-/-} jejunal epithelium were enriched for GO categories related to lipid and small-molecule metabolism, and this signature persisted and was further amplified under HFD challenge, indicating enhanced sensitivity to luminal lipid load. Because enterocyte lipid uptake culminates in cytosolic lipid droplet (LD) storage before export as chylomicrons, we profiled a curated set of 116 LD-associated genes (**Beilstein et al., 2016**) to examine whether NK2R modulates LD dynamics. In *Tacr2*^{-/-} males, hierarchical clustering segregated HFD-fed *Tacr2*^{-/-} samples from all other groups, whereas wild-type (WT) controls clustered together irrespective of diet (**Figure 4A**). *Tacr2*^{-/-} females also exhibited broad up-regulation of LD genes, although clustering by genotype and diet was less pronounced (**Figure 4—figure supplement 2A**).

The enrichment of lipid-metabolic pathways and coordinated upregulation of the enterocyte LD program in the *Tacr2*^{-/-} jejunal epithelium suggested that NK2R constrains the absorptive phase of dietary lipid handling and, therefore, regulates luminal lipid uptake. Following oil gavage, *Tacr2*^{-/-} mice exhibited normal gastrointestinal transit (**Figure 4—figure supplement 1A**) but displayed a higher post-prandial triglyceride excursion (oral lipid-tolerance test, OLTT) and increased jejunal lipid retention 4 hr after dosing in both sexes (**Figure 4B–H**). Importantly, the OLTT reports the appearance of intestinally derived triglycerides in plasma. Thus, the enhanced excursion in *Tacr2*^{-/-} mice indicates increased intestinal lipid entry into the circulation. The concurrent increase in jejunal lipid stores suggests that luminal uptake and intracellular storage of lipids are augmented, consistent with the observed induction of the LD program. Furthermore, pharmacological modulation recapitulated these effects: in diet-induced obese mice pre-treated with tyloxapol to inhibit lipoprotein lipase (LPL) and to block peripheral triglyceride clearance, the NK2R antagonist GR159897 elevated chylomicron/very-low-density lipoprotein (VLDL) triglycerides, whereas the NK2R agonist GR64349 produced the opposite outcome (**Figure 4I**). Chronic HFD feeding for 12 weeks markedly expanded intestinal epithelial LD stores in *Tacr2*^{-/-} mice of both sexes relative to wild-type controls (~ twofold in males, ~ fivefold in females; **Figure 4J**) without altering food intake, intestinal motility, metabolic rates, or locomotor activity (**Figure 4—figure supplement 1B–K, Figure 4—figure supplement 2B–K**). Together, these data support a model in which NK2R signaling tonically regulates lipid metabolism in enterocytes by tuning luminal lipid uptake, intracellular LD storage, and chylomicron secretion.

Since adipose tissue is one of the major depots for chylomicron-derived fatty acids (**Bickerton et al., 2007**), the intestinal phenotypes above predict altered lipid delivery to fat depots. Moreover, *Tacr2* expression in white adipose tissue (WAT) (**Figure 4—figure supplement 3A**) raises the possibility of adipose-autonomous NK2R actions independent of the intestine. We, therefore, asked whether loss of NK2R remodels adipose transcriptional programs and wished to distinguish secondary consequences of increased intestinal export from adipose-intrinsic NK2R effects. *Tacr2* is expressed in WAT, with comparable expression in both adipocyte and mesenchymal cell fractions (**Figure 4—figure supplement 3A**). To assess potential transcriptional changes, we performed RNA sequencing with

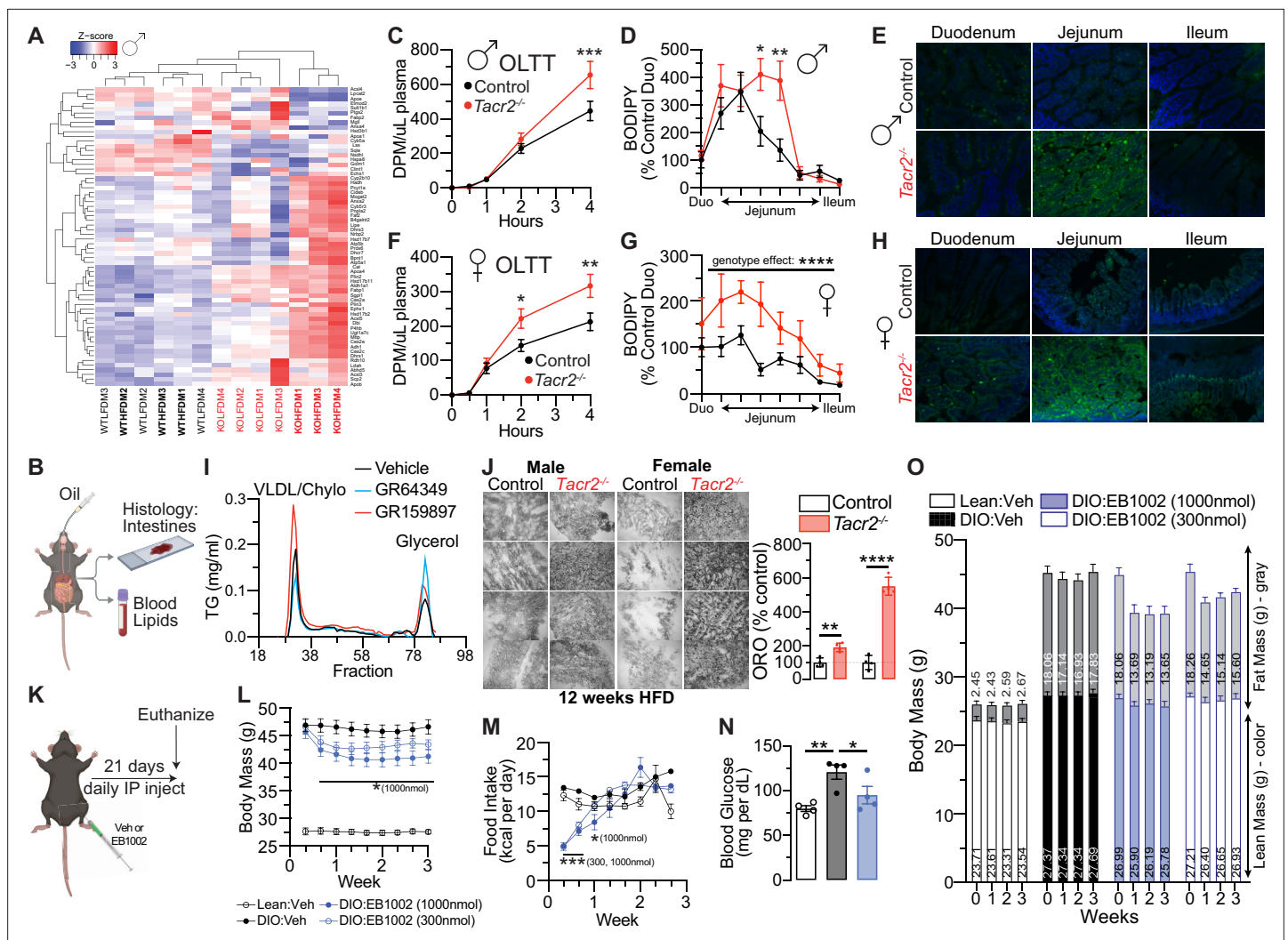


Figure 4. *Tacr2*^{-/-} mice exhibit enhanced intestinal lipid absorption. (A) Heat maps display Z-scores of normalized expression levels for genes encoding lipid droplet-associated proteins in male mice across experimental conditions. Hierarchical clustering of individual samples (top dendrogram) and genes (left dendrogram) is shown, with branch lengths reflecting the similarity between clusters. Wild-type control (WT) samples are labeled in black text, *Tacr2*^{-/-} (KO) samples in red text, and samples from mice on a high-fat diet (HFD) are indicated in bold. (B) Schematic representation of the experimental workflow for assessing intestinal lipid absorption (created with BioRender.com). Mice were administered an oral gavage of oil containing either ¹⁴C-triolein (C, F) or C16-BODIPY (D, E, G, H), followed by quantification of lipids in intestinal tissue and blood. (C, F) Oral lipid tolerance test (OLTT) was performed in male (C) and female (F) control and *Tacr2*^{-/-} mice. Blood lipid levels were measured over a 4 hr period following oral gavage and expressed as disintegrations per minute (DPM) per μ L of plasma. n=8 control males, n=7 *Tacr2*^{-/-} males for (C); n=7 control females, n=9 *Tacr2*^{-/-} females for (F). (D, G) BODIPY fluorescence was measured in intestinal sections from male (D) and female (G) control and *Tacr2*^{-/-} mice following oral gavage with oil containing C16-BODIPY. n=5 control males, n=5 *Tacr2*^{-/-} males (D); n=6 control females, n=7 *Tacr2*^{-/-} females (G). (E, H) Representative fluorescence images of intestinal sections from male (E) and female (H) mice corresponding to the samples shown in panels (D) and (G), respectively. (I) Triglyceride levels were measured in each plasma fraction from wild-type mice fed a high-fat diet (HFD) for 10 weeks and pretreated with low-fat diet (LPL) inhibitor and either vehicle, the neurokinin-2 receptor (NK2R) agonist GR64349, or NK2R antagonist GR159897 prior to oral oil gavage. n=8 mice per treatment group. (J) Representative images and quantification of Oil Red O (ORO)-stained intestinal sections from male and female control and *Tacr2*^{-/-} mice following 12 weeks of HFD feeding. n=4 control males, n=4 *Tacr2*^{-/-} males, n=4 control females, n=4 *Tacr2*^{-/-} females. (K) Schematic representation of the experimental workflow for assessing the effects of NK2R agonism on body mass (created with BioRender.com). Diet-induced obese (DIO) wild-type mice were administered daily intraperitoneal injections of EB1002 or vehicle for 21 consecutive days. (L, M) Body weights (L) and food intake (M) were monitored over a 3 week period in lean and diet-induced obese (DIO) mice treated with vehicle, high-dose EB1002 (1000 nmol), or low-dose EB1002 (300 nmol). n=8 (lean: vehicle), n=7 (DIO: vehicle), n=8 (DIO: EB1002, 1000 nmol) and n=8 (DIO: EB1002, 300 nmol). (N) The fasted blood glucose levels were determined in lean and diet-induced obese (DIO) mice treated with vehicle or high-dose EB1002 (1000 nmol). n=4 (lean: vehicle), n=4 (DIO: vehicle), and n=4 (DIO: EB1002, 1000 nmol). (O) Lean mass and fat mass were measured over 3 weeks in lean and diet-induced obese (DIO) mice treated with vehicle, high-dose EB1002 (1000 nmol), or low-dose EB1002 (300 nmol). n=8 (lean: vehicle), n=7 (DIO: vehicle), n=8 (DIO: EB1002, 1000 nmol) and n=8 (DIO: EB1002, 300 nmol). Statistical analyses were performed using two-way ANOVA with repeated measures (C, D, F, G, L, M), an unpaired two-

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Figure 4 continued

tailed Student's t-test (**B, J**), one-way ANOVA with Holm-Sidak post hoc analysis (**N**), or two-way ANOVA with Holm-Sidak post hoc analysis (**O**) Data are presented as mean $\pm$ SEM. **** $p < 0.0001$, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ^{ns} $p > 0.05$. Black bars/lines indicate control mice, red bars/lines indicate *Tacr2*^{-/-} mice, and blue bars/lines indicate NK2R agonist-treated mice.

The online version of this article includes the following figure supplement(s) for figure 4:

Figure supplement 1. Male *Tacr2*^{-/-} mice exhibit comparable gross phenotypes and feeding behaviors to control mice during chronic high-fat diet (HFD) feeding.

Figure supplement 2. Female *Tacr2*^{-/-} mice exhibit comparable gross phenotypes and feeding behaviors to control mice during chronic high-fat diet (HFD) feeding.

Figure supplement 3. *Tacr2*^{-/-} mice show similar gene expression profiles in perigonadal white adipose tissues.

perigonadal WAT at early (1 week) and chronic (12 weeks) time points after HFD exposure. (**Figure 4—figure supplement 3B**). Remarkably, the PCA revealed no genotype-dependent clustering in either male or female mice at either time point (**Figure 4—figure supplement 3C–J**). Furthermore, the number of differentially expressed genes between *Tacr2*^{-/-} and wild-type mice was negligible, indicating that *Tacr2* deletion does not appreciably alter the WAT transcriptome under HFD conditions. Together, these results suggest that, despite robust intestine-specific effects on lipid metabolism, the absence of NK2R has a limited impact on adipose transcriptional remodeling during diet-induced obesity.

Given that acute pharmacological activation of NK2R lowered chylomicron-associated triglycerides, we next asked whether sustained NK2R agonism alters whole-body energy balance. Diet-induced obese (DIO) mice received daily intraperitoneal injections of the NK2R-selective peptide agonist EB1002 (1000 nmol per kg) for 21 days (**Figure 4K**). EB1002 is a long-acting NK2R agonist recently reported by the Gerhart-Hines group to improve cardiometabolic parameters in DIO models (**Sass et al., 2024**). In our experiments, EB1002 treatment reduced body weight, perigonadal fat mass, and fasting glycemia of DIO mice relative to vehicle controls, concordant with the recent report (**Sass et al., 2024; Figure 4L, N and O**). Food intake was transiently suppressed during the first week of treatment but normalized thereafter (**Figure 4M**), indicating that the weight loss was not solely attributable to persistent hypophagia. Together with the acute lipid-handling phenotypes, these data position NK2R as a key direct regulator of intestinal lipid mobilization and long-term systemic energy balance.

Male *Tacr2*^{-/-} mice have altered microbial populations

To determine whether *Tacr2*^{-/-} males are uniquely prone to HFD-induced dysbiosis, fecal bacterial communities were profiled by 16 S rRNA gene sequencing. α -diversity, assessed by the Shannon index, was reduced in *Tacr2*^{-/-} mice irrespective of diet (**Figure 5A**). β -diversity analysis (weighted UniFrac distances) revealed significant main effects of genotype (PERMANOVA, $p = 0.002$) and diet ($p = 0.017$), as well as a genotype-diet interaction ($p = 0.001$) (**Figure 5B**), indicating that both factors independently and synergistically shape the fecal microbiota, with *Tacr2* deletion predisposing to larger diet-driven compositional shifts.

At the phylum level, HFD feeding increased the relative abundance of *Firmicutes* and concomitantly decreased *Bacteroidetes*, yielding the characteristic rise in the *Firmicutes*:*Bacteroidetes* ratio (**Lee et al., 2017; Vaughn et al., 2017; Crawford et al., 2019; Figure 5C**). A similar shift was observed in *Tacr2*^{-/-} mice under HFD conditions. Linear discriminant analysis effect size (LEfSe) identified 66 operational taxonomic units (OTUs) that were diet-responsive and 104 that were genotype-responsive (**Figure 5D and E**); 18 OTUs were influenced by both variables (**Figure 5F**). Notably, OTU 215 was completely absent in *Tacr2*^{-/-} mice, revealing a genotype-specific microbial deficit.

Discussion

In this report, we identify neurokinin-2 receptor (NK2R) signaling as a nexus between intestinal lipid handling, mucosal immunity, and host-microbe interactions. *Tacr2* deletion or acute pharmacological antagonism augmented chylomicron export and massively expanded lipid-droplet (LD) stores in jejunal enterocytes, whereas NK2R agonism suppressed postprandial triglyceridemia and reduced

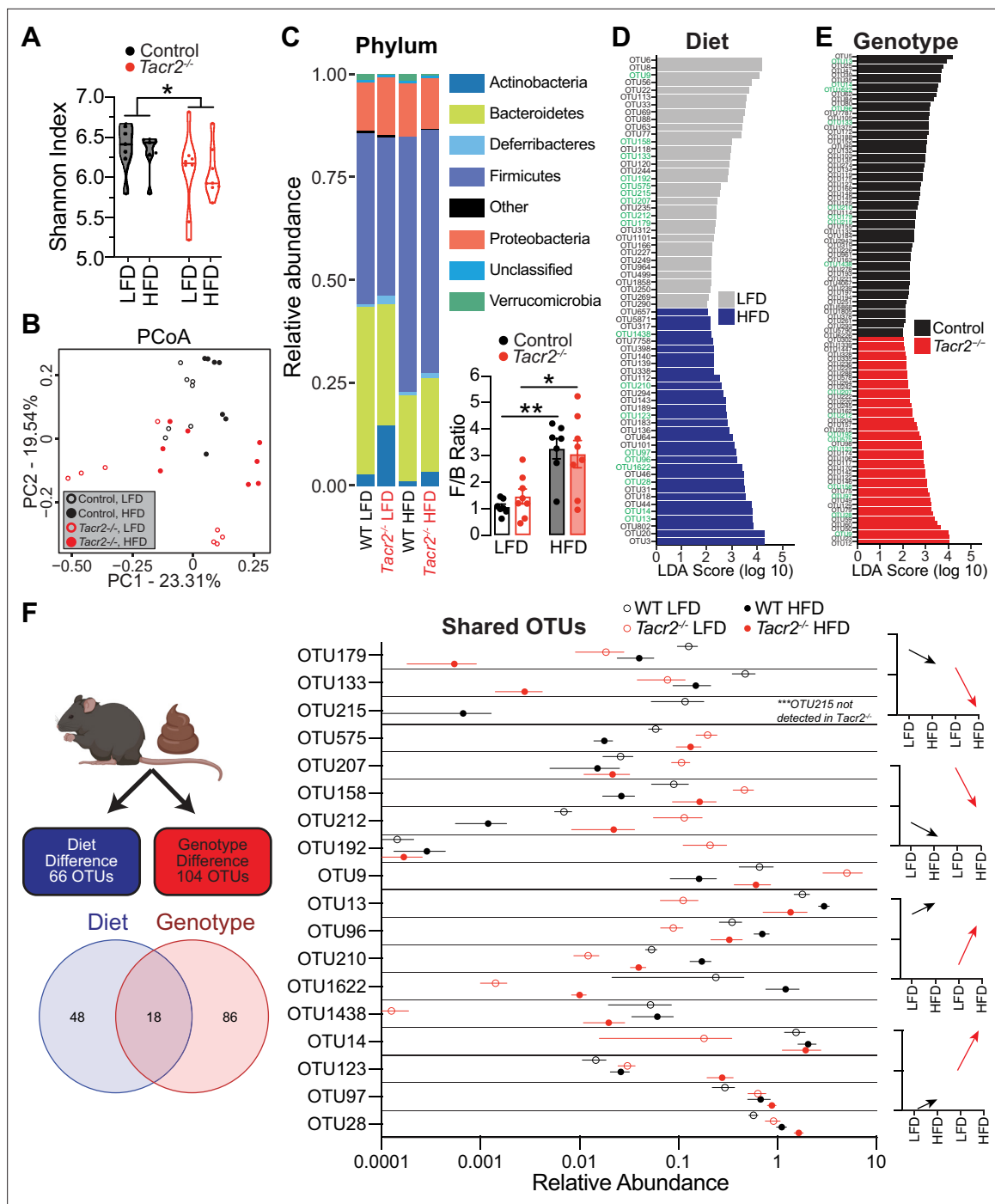


Figure 5. 16 S rRNA sequencing analysis of gut microbial populations. **(A)** Shannon index (α -diversity) of fecal microbial populations in control and *Tacr2^{-/-}* mice fed a low-fat diet (LFD) or high-fat diet (HFD). $n=7$ control (LFD), $n=7$ control (HFD), $n=8$ *Tacr2^{-/-}* (LFD), $n=8$ *Tacr2^{-/-}* (HFD). Data were analyzed using two-way ANOVA followed by Holm–Sidak’s post hoc analysis; genotype effect: $p=0.039$, diet effect: ns, genotype $\times$ diet interaction: ns. **(B)** Principal Coordinate Analysis (PCoA) plot based on weighted UniFrac distances illustrating β -diversity of microbial communities in *Tacr2^{-/-}* and control mice fed a LFD or HFD. Each point represents a microbial community from an individual sample, color-coded by genotype (*Tacr2^{-/-}*, red; control, black) and diet (LFD, open circles; HFD, closed circles). **(C)** Microbial populations were sorted at the phylum level across experimental conditions. Firmicutes/Bacteroidetes (F/B) ratio was calculated in control and *Tacr2^{-/-}* mice fed a LFD or HFD. $n=7$ control (LFD), $n=8$ *Tacr2^{-/-}* (LFD), $n=7$ control (HFD), $n=8$ *Tacr2^{-/-}* (HFD). Data were analyzed using two-way ANOVA followed by Holm–Sidak post hoc analysis. Data are shown as mean $\pm$ SEM; ** $p<0.01$, * $p<0.05$. **(D, E)** Operational Taxonomic Units (OTUs) were analyzed using Linear Discriminant Effect Size (LEfSe) to identify taxa differentially enriched by diet or genotype. **(D)** Bar graph showing Linear Discriminant Analysis (LDA) scores for 66 diet-dependent OTUs (34 associated with LFD, 32 associated with HFD). **(E)** Bar graph showing LDA scores for 104 genotype-dependent OTUs (60 associated with control mice, 44 associated in *Tacr2^{-/-}* mice). **(F)**

Figure 5 continued on next page

Figure 5 continued

Relative abundances of 18 OTUs shared between the diet- and genotype-dependent analyses (created with BioRender.com). OTUs were filtered using a Linear Discriminant Analysis (LDA) score threshold >2.0. In panels (D) and (E), OTUs shared between diet and genotype analyses are indicated in green text.

adiposity. These bidirectional effects suggest that NK2R serves as a molecular switch for intestinal lipid mobilization. The downstream cascade by which NK2R transduces neuropeptidergic input to LD dynamics remains to be elucidated; however, transcriptomic profiling of *Tacr2*^{-/-} jejunal epithelium from the present study uncovered the coordinated up-regulation of a core LD gene module, suggesting that tonic NK2R activity normally restrains this program.

Bisque deconvolution revealed that *Tacr2*-null males, but not females, remodeled goblet and tuft cell lineages when challenged with a HFD. These secretory cell types shape barrier function and epithelial sensing of luminal antigens (Gustafsson and Johansson, 2022; Feng et al., 2024) and are associated with intestinal inflammatory responses (Steele et al., 2016; Nowarski et al., 2015; Nyström et al., 2021; Yi et al., 2019), providing a plausible link to the striking male-specific protection from DSS-induced colitis. The absence of similar lineage shifts in females may underlie their failure to gain colitis resistance despite harboring ever larger epithelial lipid stores. Sex hormones, which modulate tachykinin and receptor expression (Villablanca and Hanley, 1997; Pinto et al., 2009), may be one contributing factor to this divergence and warrant further study.

Gut-innervating nociceptors regulate mucosal protection against dietary, microbial, and inflammatory insults (Zhang et al., 2022; Lai et al., 2020; Di Giovangiulio et al., 2015). These nociceptors control mucin secretion from goblet cells to create a physical barrier between the epithelium and the luminal contents (Yang et al., 2022). Recent evidence suggests that nociceptor-derived Substance P plays a crucial role in regulating mucosal protection during colitis and dysbiosis (Zhang et al., 2022). *Tac1* excision (which encodes both Substance P and NKA, the endogenous ligand for NK2R) from these neurons aggravates colitis (Zhang et al., 2022), whereas global NK2R ablation confers male-biased protection, suggesting that proinflammatory signaling through NK2R on immune or stromal cells may outweigh epithelial loss of tachykinin responsiveness during injury. In future efforts, conditional *Tacr2* deletion in distinct tissues as well as intestinal epithelial versus hematopoietic compartments will be essential to define the cell- and tissue-specific roles of NK2R.

Consistent with previous studies (Crawford et al., 2019), the HFD increased the fecal *Firmicutes*:*Bacteroidetes* (F:B) ratio after only one week, and this shift was reproduced in *Tacr2*^{-/-} mice. We note here that our diets are matched for nutritional composition, differing only in fat and sugar sources and percent of total %kcal, limiting confounding effects of dietary fiber in our study. 18 operational OTUs were jointly influenced by genotype and diet; 12 belonged to the *Clostridiales* order [Families: *Lachnospiraceae* (5), *Ruminococcaceae* (3), and unclassified lineages (4)]. Members of *Lachnospiraceae* have been linked to constipation, whereas *Ruminococcaceae* abundance correlates with irritable bowel syndrome and type 2 diabetes (Jackson et al., 2018). Notably, the HFD expanded *Ruminococcus gnavus* (OTU575) and depleted *Akkermansia muciniphila* (OTU 1438), a combination associated with impaired barrier function through altered immune tolerance, short-chain fatty acid production, and mucin layer maintenance (Everard et al., 2013; Yan et al., 2021; Henke et al., 2019). Conversely, *Bifidobacterium pseudolongum* (OTU 9), which is often negatively associated with metabolic and inflammatory disorders (Tatsuoka et al., 2023; Ma et al., 2023; Bo et al., 2020), was increased. Although this composite profile resembles that observed in inflammatory bowel disease, *Tacr2* deletion conferred resistance to DSS-induced colitis in males. These findings underscore the complexity of host-microbiota interactions and suggest that NK2R signaling modulates intestinal inflammation through mechanisms that extend beyond simple taxonomic shifts, potentially involving neuropeptide-dependent regulation of epithelial and immune function.

Collectively, our findings position NK2R at the intersection of metabolic and inflammatory pathways in the intestine, supporting the tachykinin-NK2R axis as highly tractable to therapeutic intervention. In principle, NK2R agonists could be leveraged to treat metabolic diseases, including obesity, whereas NK2R antagonists may ameliorate mucosal inflammation in inflammatory bowel disease. The pronounced sexual dimorphism underscores the need to consider sex as a biological variable in the development of NK2R-targeted therapeutics. Based on our findings, we expect that delineating the precise mechanisms by which NK2R integrates neuronal, immune, and microbial cues to regulate lipid

droplet dynamics and mucosal immunity in the intestine will yield important mechanistic insights and advance the development of targeted therapies for metabolic and inflammatory diseases.

Methods

Animals

Male and female C57BL/6 J mice (8–12 weeks old) were group-housed under a 12 hr light/dark cycle with free access to food and water. Animals were fed either a low-fat control diet (CD; D14042701, Research Diets, New Brunswick, NJ; 73% kcal carbohydrate, 10% kcal fat, 17% kcal protein) or a Western-style high-fat diet (HFD; D12079B, Research Diets; 43% kcal carbohydrate, 40% kcal fat, 17% kcal protein). Body weights were measured weekly between 09:00 and 11:00 hr. All animal protocols were reviewed and approved by the Institutional Animal Care and Use Committee (IACUC) at The Scripps Research Institute (IACUC Protocol Number: 18–0016).

RNA extraction and RT-qPCR analysis

Mice were euthanized between 09:00 and 11:00 hr, and tissues were promptly collected and washed with ice-cold phosphate-buffered saline (PBS, 4 °C) before snap-freezing in liquid nitrogen. The small intestine was washed with chilled PBS, opened longitudinally on ice, and the contents were removed. Intestinal epithelium was isolated by scraping with glass slides from the underlying submucosal layer and snap-frozen in liquid nitrogen. Samples were stored at –80 °C until processing. Tissues were homogenized in TRIzol reagent, followed by phase separation with 1-bromo-3-chloropropane via centrifugation (8000 g, 4 °C). The upper aqueous phase was collected, and total RNA was purified using an RNeasy Mini Kit (Cat. No. 74104, Qiagen, Redwood City, CA) per the manufacturer's instructions. cDNA synthesis was performed using iScript Reverse Transcription Supermix (Cat. No. 1708840, Bio-Rad, Hercules, CA). Quantitative PCR was conducted using SsoAdvanced Universal SYBR Green Supermix (Bio-Rad) according to the manufacturer's protocols. Gene expression was normalized to reference housekeeping genes (*Actb*, *Rplp0*, or *Hprt*). RNA integrity was preserved by sanitizing all surfaces with 70% ethanol and RNase inhibitor (RNase OUT, G-Biosciences, St. Louis, MO). Data normalization was performed against either the small intestine or *Vil1* expression. Primer sequences are listed in **Supplementary file 1**.

Immunohistochemistry

The proximal small intestine was excised and flushed with ice-cold modified Bouin's fixative, then fixed in 10% buffered formalin for 24 hr at room temperature. Tissue was embedded and frozen in optimal cutting temperature (OCT) compound (Fisher Healthcare, Chino, CA) on dry ice. Sections (16 µm) were cut using a cryostat (Leica), mounted onto charged glass slides, permeabilized with 0.025% Triton X-100 in TBS, and blocked with 10% normal donkey serum and 1% bovine serum albumin (BSA) in TBST (Millipore Sigma, Cat. No. 566460). Sections were incubated with primary rabbit antibody against Neurokinin 2 Receptor (NK2R; 20 µg/mL; Cat. No. ATR-002, RRID:AB_2341078, Alomone Labs, Jerusalem, Israel), followed by incubation with Alexa Fluor 488-conjugated donkey anti-rabbit secondary antibody (Cat. No. A-21206, RRID:AB_2535792, Thermo Fisher Scientific). Sections were washed, mounted with ProLong Gold Antifade reagent containing DAPI (Thermo Fisher Scientific) for nuclear staining, and imaged using a Nikon A1 confocal microscope with a 60 x objective at room temperature. Image processing was conducted using ImageJ software version 2.0.0 (NIH, Bethesda, MD).

CRISPR-Cas9 gene editing

Two single-guide RNAs (sgRNAs) were designed to target intronic regions flanking exon 2 of the *Tacr2* gene. A single-stranded DNA (ssDNA) repair template (721 nucleotides) containing homology arms (80 nucleotides each) flanking exon 2 and a LoxP sequence within the intronic regions was synthesized. Following Cas9-mediated DNA cleavage and homologous recombination using the ssDNA repair template, loxP sites were integrated flanking exon 2 to generate *Tacr2^{fl/fl}* mice. Non-homologous end joining after Cas9 cleavage resulted in the deletion of exon 2, generating *Tacr2^{-/-}* mice. Constructs (1 ng/µL) were microinjected into fertilized eggs to produce genetically modified mice, which were validated by sequencing. Genotyping primer sequences are listed in **Supplementary file 2**.

Glucose tolerance test

Mice were fasted for 12 hr prior to glucose tolerance testing. Fasted mice received an intraperitoneal injection of glucose solution (10% w/v) at a dose of 1 g glucose per kg body weight. Blood glucose concentrations were measured from tail blood samples at 0, 15, 30, 45, 60, and 120 min post-injection using a commercial handheld glucometer.

RNA sequencing and analysis

RNA from intestinal epithelium was extracted as described above. Total RNA samples were prepared into RNA-Seq libraries using the NEBNext Ultra II Directional RNA Library Prep Kit for Illumina following the manufacturer's recommended protocol. Briefly, for each sample, 200 ng total RNA was polyA selected, and converted to double-stranded cDNA followed by fragmentation and ligation of sequencing adapters. The libraries were then PCR amplified 12 cycles using barcoded PCR primers, purified, and size-selected using AMPure XP Beads before loading onto an Illumina NextSeq 2000 for 100-base single-read sequencing. Raw sequencing reads were quality-assessed using FastQC, and adapters were removed using Cutadapt. Trimmed reads were aligned to the mouse reference genome (ENSEMBL GRCm38) using STAR aligner version 2.6.1d (Dobin *et al.*, 2013). The distribution of mapped reads across genomic features was analyzed with RSeQC v3.0.1. Differential gene expression analysis was performed using DESeq2 (v2.11.40.8) implemented in R (v4.2.2) via the Galaxy platform. Genes were considered significantly differentially expressed at thresholds of adjusted $p < 0.05$ and absolute $\log_2$ fold-change > 1.0 . Gene ontology, pathway enrichment, and network analyses were conducted using ShinyGO v0.80 (Ge *et al.*, 2020). Cell-type-specific deconvolution was performed using BisqueRNA v1.0.5 against a published single-cell RNA sequencing dataset profiling mouse small intestinal epithelium (Haber *et al.*, 2017; Jew *et al.*, 1971). RNA sequencing data were deposited in Gene Expression Omnibus (GEO) (GSE318405).

Radioactive tracer oil challenges

Mice were fasted overnight prior to testing. Fasted mice were pretreated with Poloxamer-407. 10 g of Poloxamer-407 were resuspended in 100 mL of 0.9% NaCl saline and stirred overnight at 4 °C. 10 μ L/g of body weight was administered by intraperitoneal injection right before the oil gavage with 3 μ Ci [3 H] triolein. Blood was collected at time 0, 1, 2, and 4 hr, and the plasma was separated by centrifugation. Radioactivity was measured by scintillation.

C₁₆-BODIPY absorption assay

Mice were fasted overnight before receiving an oral gavage of olive oil containing 2 μ g/g body weight of fluorescently labeled fatty acid analog C₁₆-BODIPY (4,4-difluoro-5,7-dimethyl-4-bora-3a,4a-diaza-s-indacene-3-hexadecanoic acid; Thermo Fisher, cat. no. D3821) administered at a volume of 10 μ L/g body weight. Mice were euthanized 4 hr post-gavage, and intestinal tissues were collected, snap-frozen, sectioned, and imaged as previously described (Zhang *et al.*, 2019).

Gastrointestinal transit assay

Gastrointestinal transit time was assessed by oral gavage of a semi-liquid Evans Blue dye suspension (5% Evans Blue, 0.5% methylcellulose in phosphate-buffered saline; 100 μ L per mouse). Following gavage, mice were monitored individually, and the time to appearance of the first blue-colored fecal pellet was recorded. Measurements were conducted in mice maintained on a control diet or after 12 weeks on a high-fat diet (HFD), as previously described (Anitha *et al.*, 2016).

FPLC assay

Mice were fasted overnight prior to testing and pretreated with either lipoprotein lipase (LPL) inhibitor Tyloxapol (500 μ g/g body weight), vehicle (PBS), the NK2R agonist GR64349 (10 mg/kg), or the NK2R antagonist GR159897 (10 mg/kg). Following pretreatment, mice received an oral gavage of olive oil. Blood plasma was collected 2 h post-gavage, separated by fast protein liquid chromatography (FPLC), and triglyceride levels were quantified across collected fractions.

Fecal lipid extraction

Mice were housed in wire-bottom cages for 12 hr to collect fecal pellets. Pellets were dried, weighed, and pulverized using a mortar and pestle, then incubated in a 2:1 (v/v) chloroform:methanol solution.

Samples were vortexed for 1 min and centrifuged at 1000 g for 10 min. The lower organic phase was collected, dried, and the residual lipids were weighed. Lipid content was normalized to the initial dried fecal weight (*Kraus et al., 2015*).

Feeding behavior (CLAMS)

Indirect calorimetry was performed using a computer-controlled, open-circuit system (Oxymax System) integrated into the Comprehensive Lab Animal Monitoring System (CLAMS; Columbus Instruments, Columbus, OH) (*Chang et al., 2012; Yang et al., 2009*). Single-housed, acclimated mice were placed into clear respiratory chambers (20×10×12.5 cm) equipped with a water sipper tube, a food tray connected to a balance for continuous food intake measurement, and 16 photobeams arranged in two axes at 0.5-inch intervals to monitor motor activity. Room air was circulated through each chamber at a flow rate of 0.5 L/min, and exhaust air was sampled every 15 min for 1 min. Oxygen consumption (VO₂) and carbon dioxide production (VCO₂) were measured using O₂ and CO₂ sensors (Columbus Instruments), and respiratory exchange ratio (RER) was calculated as VCO₂/VO₂. All metabolic parameters were normalized to lean body mass, which was determined by EchoMRI analysis.

Dextran sulfate sodium (DSS) colitis model

Colitis was induced by providing mice with ad libitum access to drinking water containing dextran sulfate sodium (DSS; 3.0% w/v for males, 5.0% w/v for females; MP Biomedicals, cat. no. MFCD00081551) for seven consecutive days. Mice were monitored daily for changes in body weight and clinical condition. Animals that lost more than 25% of their initial body weight were euthanized and excluded from further analysis. On day 7, all remaining mice were euthanized, and tissues were collected for downstream analyses (*Chassaing et al., 2014*).

Tissue scoring parameters

The intact colon was collected, flushed with chilled phosphate-buffered saline (PBS), and fixed overnight in 4% paraformaldehyde (PFA) in PBS. The distal 1 cm segment was cryosectioned, stained with hematoxylin and eosin (H&E), and imaged. Histological sections were scored based on epithelial integrity (0–4) and immune cell infiltration (0–4) for a maximum aggregate score of 0–8 (*Erben et al., 2014*). Epithelial integrity was scored as follows: 0, normal morphology; 1, loss of goblet cells; 2, loss of goblet cells in multiple regions; 3, loss of crypts; 4, loss of crypts in multiple regions. Immune cell infiltration was scored as follows: 0, no infiltration; 1, infiltration around the crypt base; 2, infiltration reaching the lamina muscularis; 3, extensive infiltration of the lamina muscularis with mucosal thickening and edema; 4, infiltration extending into the submucosa.

Fecal protein extraction for Lipocalin-2 quantification

Fecal pellets were collected at the time of sacrifice and flash-frozen in liquid nitrogen. Approximately 100 mg of frozen feces per sample was weighed, suspended in 1 mL of 1% Tween-20 in PBS, and vortexed vigorously for 10 min. Samples were then centrifuged at 15,000 g for 10 min at 4 °C. The resulting supernatant was collected and diluted (1:50 for control samples, 1:2000 for DSS-treated samples) prior to measurement of Lipocalin-2 concentrations using a commercial mouse Lipocalin-2 ELISA kit (R&D Systems, cat. no. DY1857) (*Chassaing et al., 2012*).

Intestinal permeability assay

Mice were fasted beginning at 09:00 for 4 hr prior to oral gavage with fluorescein isothiocyanate-dextran 4000 (FD4; Millipore Sigma, cat. no. 46944) at a dose of 0.6 mg/g body weight from a 100 mg/mL stock solution. Four hours after gavage, blood was collected via cardiac puncture into BD Microtainer tubes containing lithium heparin and maintained on ice. Plasma was isolated by centrifugation at 5000 g for 5 min at 4 °C. Plasma from fasted mice that did not receive FD4 was used to generate a standard curve (0–10 µg/mL FD4), and plasma FD4 concentrations in experimental samples were quantified accordingly (*Pedicord et al., 2016; Woting and Blaut, 2018*).

Microbiome 16S rRNA sequencing and microbial analysis

Fecal pellets were collected from mice, and genomic bacterial DNA was extracted using the DNeasy PowerLyzer PowerSoil Kit (Qiagen, cat. no. 12855) according to the manufacturer's instructions. DNA

concentration was quantified and samples were submitted to GENEWIZ (South Plainfield, NJ, USA) for 16S-EZ sequencing using Illumina paired-end 2×250 bp methodology. Raw sequence reads were processed with Cutadapt (v1.9.1) to remove adapter sequences and assigned to operational taxonomic units (OTUs) based on a 97% similarity threshold using VSEARCH (v1.9.1) and QIIME (v1.9.1). Differential microbial abundance analyses were performed using linear discriminant analysis effect size (LEfSe) (Segata et al., 2011).

Statistical analysis

Sample sizes were informed by relevant published studies using comparable assays and pilot experiments assessing phenotype magnitude, with the aim of detecting statistically significant differences while minimizing animal use. Experimental groups were defined by genotype and sex, and genotypes were confirmed by PCR. Within each genotype and sex, animals were randomly selected for assays. Investigators performed blinded scoring whenever feasible. Data are presented as mean ± SEM, and the number of biological replicates is provided in the corresponding figure legends. All experiments were repeated at least three times. Comparisons between two groups were performed using an unpaired two-tailed Student’s t-tests. Repeated-measures two-way ANOVA was used for comparisons across time points. Analyses involving more than two groups were performed using one-way or two-way ANOVA followed by Sidak’s or Tukey’s post hoc multiple-comparison tests, as appropriate. Differences in UniFrac distances were assessed using PERMANOVA. Statistical significance was defined as p<0.05. Outliers were identified using Grubbs’ test. Statistical analyses were performed using GraphPad Prism v10.3.

Materials availability statement

The mouse lines generated in this study, including *Tacr2*^{-/-} and *Tacr2*^{fl/fl}, are available from the corresponding author upon reasonable request.

Acknowledgements

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Author contributions

Pedro A Perez, Chung-Chih Liu, Conceptualization, Data curation, Formal analysis, Funding acquisition, Validation, Investigation, Visualization, Methodology, Writing – original draft, Writing – review and editing; Alessandra Ferrari, Conceptualization, Data curation, Formal analysis, Validation, Investigation, Visualization, Methodology, Writing – original draft, Writing – review and editing; Nicole K Littlejohn, Conceptualization, Data curation, Formal analysis, Investigation, Visualization, Methodology; John Paul Kennelly, Emma Marie Robinson, Data curation, Formal analysis, Investigation, Visualization, Methodology; Vân TB Nguyen-Tran, Jon Athanacio, Data curation; Sean B Joesph, Conceptualization; Zaid Amso, Resources, Supervision, Funding acquisition, Project administration; Peter Tontonoz, Conceptualization, Resources, Supervision, Funding acquisition, Investigation, Methodology, Project administration, Writing – review and editing; Supriya Srinivasan, Conceptualization, Resources, Supervision, Funding acquisition, Investigation, Methodology, Writing – original draft, Project administration, Writing – review and editing

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Ethics

All animal protocols (#18-0016) were reviewed and approved by the Institutional Animal Care and Use Committee (IACUC) at The Scripps Research Institute.

Peer review material

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Additional files

Supplementary files

Supplementary file 1. qPCR primers used in this study. Forward and reverse primer sequences used for quantitative PCR (qPCR) analyses. All primers were obtained from Integrated DNA Technologies (IDT).

Supplementary file 2. Genotyping primers used in this study. Forward and reverse primer sequences used for genotyping *Tacr2*-null and floxed alleles. All primers were obtained from Integrated DNA Technologies (IDT).

Source data 1. Numerical data underlying the figures and figure supplements. Numerical source data are provided in a single Excel workbook, with separate worksheets labeled according to the corresponding figure and panel. Each worksheet contains the associated data and experimental group labels.

MDAR checklist

Data availability

Numerical data underlying the figures and figure supplements are provided in **Source data 1**. Bulk RNA-sequencing data generated in this study have been deposited in the Gene Expression Omnibus (GEO) under accession number GSE318405. The published mouse small-intestinal single-cell RNA-sequencing dataset used for cell-type deconvolution is available through GEO under accession number GSE92332. Human gene-expression data were obtained from the Human Protein Atlas.

The following dataset was generated:

Author(s)	Year	Dataset title	Dataset URL	Database and Identifier
Perez PA, Liu C, Ferrari A, Littlejohn N, Kennelly JP, Robinson EM, Nguyen-Tran VT, Athanacio J, Joseph SB, Amso Z, Tontonoz P, Srinivasan S	2026	NK2R signaling governs intestinal lipid mobilization and mucosal inflammation	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE318405	NCBI Gene Expression Omnibus, GSE318405

The following previously published dataset was used:

Author(s)	Year	Dataset title	Dataset URL	Database and Identifier
Haber AL, Biton M, Rogel N, Herbst RH, Shekhar K, Smillie C, Burgin G, Delorey TM, Howitt MR, Katz Y, Tirosch I, Beyaz S, Dionne D, Zhang M, Raychowdhury R, Garret WS, Shi HN, Rozenblatt-Rosen O, Yilmaz O, Xavier R, Regev A	2017	A single-cell survey of the small intestinal epithelium	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE92332	NCBI Gene Expression Omnibus, GSE92332

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